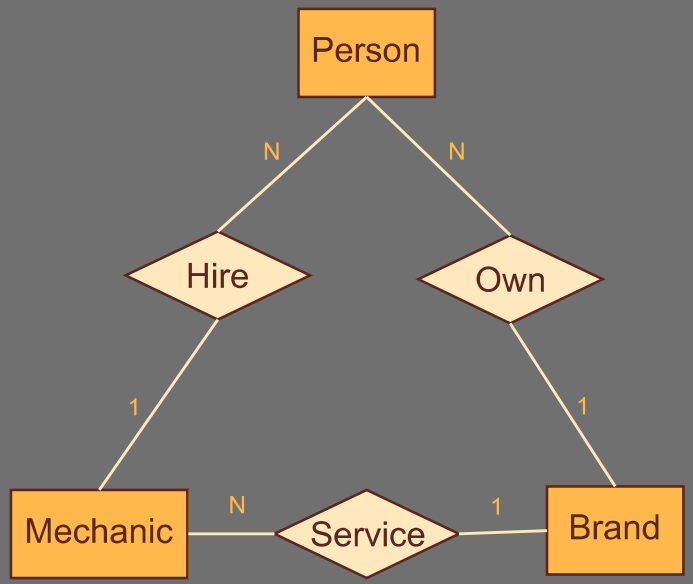
ERD Limitations

Restrictions we can get around

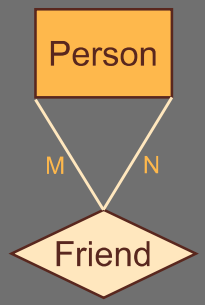
- Restrictions on cardinalities
- No flexibility on relationship keys
- Only unions in classes
- No singleton subclasses

Relationship cycles allow inconsistency



- Persephone owns a Bently.
- Persephone hires Mike.
- Mike services Buicks.

Relationship cycles allow inconsistency



- Persephone is friends with Priscilla.
- Priscilla is **not** friends with Persephone.

- ERD cannot enforce symmetry of "mutual friends".

Implementation “Fixes”

Can add restrictions via

- User-defined constraints (SQL)
- Triggers (SQL)
- Functions to check conditions (SQL/application)
- CRUD API (SQL/application)

Pragmatic Design Choice

Choose simplest, most-accurate ERD allowing desired semantics.

Impose additional restrictions in implementation.